



MASTER SOLUTION — UNIT I: INTRODUCTION TO CLOUD COMPUTING

Subject: Cloud Computing

Branch: Computer Engineering / AI & DS

Pattern: 2024

Sem: V

Total Questions: 17

Cloud Fundamentals: Definition, Importance of cloud computing, Advantages and Disadvantages of Cloud Computing, Characteristics, Categories of Clouds: Private clouds, Public clouds Cloud Service Models: SaaS, PaaS, IaaS, Cloud Architecture, Cloud Storage: Distributed Data Storage, Data management, Cloud Deployment Models

Q1 Define Cloud Computing. Explain its importance and essential characteristics as per NIST. [6M]

Ans

•

Cloud Computing is defined by the National Institute of Standards and Technology (NIST) as a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources, such as networks, servers, storage, applications and services, that can be rapidly provisioned and released with minimal management effort or service provider interaction. In simple terms, cloud computing lets an organisation rent computing power the way it rents electricity, drawing exactly as much as needed and paying only for what is actually consumed.

Importance of Cloud Computing: Cloud computing has become central to modern IT because it removes the need for organisations to invest heavily in physical infrastructure such as servers, cooling systems and network hardware. Instead of purchasing and maintaining this equipment, businesses rent computing power, storage and software from a specialised provider and pay only for what they consume, which is commonly called a pay-as-you-go model.

This shift changes IT spending from a capital expense, where money is locked into hardware that may sit idle much of the time, into an operating expense that scales directly with business activity. This is especially valuable for startups and small organisations that cannot afford large upfront investment, but it is equally important for large enterprises that want to redirect capital toward core business innovation rather than data-centre maintenance.

Cloud computing also allows applications to scale instantly during demand spikes, such as an e-commerce website during a festival sale or an online examination portal during result day, without the business having to own the extra hardware permanently. Once the peak passes, the extra capacity is released automatically and the organisation stops paying for it, which is something a traditional owned data centre can never achieve.

Essential Characteristics as per NIST: NIST identifies five essential characteristics that distinguish true cloud computing from ordinary outsourced hosting or a simple managed data centre.

- **On-Demand Self-Service:** A consumer can unilaterally provision computing capabilities, such as server time and network storage, automatically whenever needed, without requiring human

	interaction with each service provider. For example, a developer can log into the AWS Management Console and spin up a new virtual machine within minutes, entirely through a self-service interface, with no phone call or approval ticket required from the provider's staff. • Broad Network Access: Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms such as mobile phones, tablets, laptops and workstations. This allows employees to access company applications and data from any device with an internet connection, whether they are in the office, at home or travelling. • Resource Pooling: The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. The customer generally has no knowledge or control over the exact physical location of the provided resources, though they may be able to specify a broader region such as a country or a data-centre zone. • Rapid Elasticity: Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time, which supports sudden traffic surges without any manual planning. • Measured Service: Cloud systems automatically control and optimise resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service, such as storage, processing, bandwidth and active user accounts. Resource usage is monitored, controlled and reported, providing transparency for both the provider and the consumer of the utilised service, which directly feeds into the billing engine. • Multi-Tenancy: Although not one of the original five NIST characteristics, multi-tenancy is widely recognised as a defining trait of cloud computing, where a single instance of software and its supporting infrastructure serves multiple customers, each isolated logically from the others, which improves hardware utilisation and lowers the per-user cost of running the service. Together, these characteristics make cloud computing behave like a utility service, comparable to electricity or water supply, where the consumer draws exactly as much computing power as needed at a given moment and is billed strictly according to actual consumption rather than a fixed subscription for idle capacity.
--	---

Q2	List and explain the advantages and disadvantages of Cloud Computing. [5M]
Ans	Cloud computing shifts the ownership of hardware, software and specialised operating staff away from the end organisation to a specialised third-party provider. This brings clear operational and financial benefits, but it also introduces new risks and dependencies that every adopting organisation must weigh carefully before migrating critical workloads. Advantages of Cloud Computing: • Cost Reduction: Organisations avoid large capital expenditure on servers, storage arrays and data-centre facilities, paying instead on a pay-as-you-go basis, which converts fixed infrastructure costs into variable operating costs that scale directly with actual usage.

- **Scalability and Elasticity:** Resources can be scaled up or down almost instantly to match workload, so a business never over-provisions capacity for a peak that may occur only a few days a year, nor wastes money on idle capacity during low-demand periods.
- **Accessibility and Mobility:** Applications and data hosted in the cloud can be accessed from anywhere with an internet connection, on virtually any device, which supports remote work, global collaboration and business continuity during disruptions.
- **Automatic Maintenance:** The cloud provider handles patching, security updates and hardware replacement centrally, freeing the organisation's own IT staff to focus on core business applications instead of routine infrastructure upkeep.
- **Disaster Recovery:** Cloud providers replicate data across multiple geographically distributed data centres, so recovery after a local hardware failure, fire or natural disaster is faster and more reliable than a typical in-house backup arrangement.
- **Faster Deployment:** New environments, servers and services can be provisioned within minutes through a self-service console, instead of the weeks that would otherwise be required to procure, ship and install physical hardware.
- **Environmental Efficiency:** Large cloud data centres achieve much higher server utilisation and better cooling efficiency than thousands of small independent data centres, reducing the overall carbon footprint of computing per unit of useful work delivered.

Disadvantages of Cloud Computing: Despite these benefits, moving to the cloud is not free of drawbacks and requires careful evaluation before committing mission-critical workloads.

- **Dependency on Internet:** Since all cloud services are accessed over the network, poor connectivity or a complete outage directly blocks access to critical applications and data, which is a risk that does not exist for purely local, on-premises systems.
- **Data Security and Privacy:** Storing sensitive data on third-party servers raises genuine concerns about unauthorised access, data breaches and compliance with data-protection regulations such as GDPR, especially when data crosses national borders.
- **Vendor Lock-in:** Applications built using a specific provider's proprietary APIs, storage formats and managed services can become difficult, slow and costly to migrate to another provider later, effectively tying the organisation to one vendor's pricing and roadmap.
- **Limited Control:** The consumer has little visibility or control over the underlying physical infrastructure, exact patch schedules or hardware choices, which can be a serious concern for organisations with strict audit or compliance requirements.
- **Downtime Risk:** Even the largest providers occasionally suffer regional outages, and because many unrelated businesses depend on the same provider, a single outage can simultaneously affect a very large number of customers across industries.
- **Hidden Long-Term Costs:** While pay-as-you-go pricing looks attractive initially, unmonitored usage, data-transfer charges and premium support fees can accumulate over time to exceed the cost of an equivalent on-premises deployment.

Q3 Explain the seven-step model of migration into a Cloud. [6M]

replacing local file storage with object storage APIs, this rearchitecting work is carried out at this stage to make the application genuinely cloud-compatible rather than merely cloud-hosted.

- **Step 5 — Augment:** The organisation augments and exploits the cloud's specific capabilities, such as auto-scaling groups, elastic load balancing, managed databases and pay-per-use billing, in order to derive maximum operational and financial benefit from the target cloud platform, rather than simply replicating the old on-premises setup inside a virtual machine.
- **Step 6 — Test:** The migrated application is thoroughly tested using the new cloud infrastructure to validate functional correctness, performance under realistic load, data integrity and security controls before it is switched over to production traffic. Parallel running with the legacy system is common at this stage to catch discrepancies.
- **Step 7 — Iterate:** The migration process itself is optimised iteratively; every round of testing, deployment and post-migration monitoring feeds back lessons that refine the migration strategy, checklist and automation scripts used for subsequent applications in the organisation's broader migration programme.

By following this structured sequence rather than an ad-hoc lift-and-shift, an enterprise substantially reduces the risk of unplanned downtime, avoids costly compatibility surprises discovered late in the process, and ensures that the migrated application performs at least as well in the cloud as it did on the original on-premises infrastructure, while also positioning it to exploit cloud-native benefits going forward.

Importance of Following the Model: Organisations that skip steps, such as moving directly from Assess to Test without properly isolating dependencies or rearchitecting the application, frequently discover hidden failures only after the application is already serving live production traffic, which can cause serious business disruption and erode stakeholder confidence in the entire migration programme.

Because most enterprises must migrate not just one application but an entire portfolio of interconnected systems, the seven-step model is rarely applied only once. Lessons learned while migrating an early, low-risk application are deliberately fed back into the planning for later, more business-critical applications, which is precisely why Step 7, Iterate, is treated as a genuine loop back to Step 1 rather than a simple final checkpoint.

Q4	Explain Software as a Service (SaaS) with a neat diagram. [6M]
Ans •	Software as a Service is a cloud service model in which a complete, ready-to-use application is hosted by the provider and delivered to customers over the internet, typically through a web browser or a lightweight client, without the customer having to install, manage, patch or maintain any underlying infrastructure, operating system or the application code itself. Working of SaaS: The provider hosts the application, its database and all supporting infrastructure on its own servers, usually distributed across multiple data centres for reliability. Customers subscribe to the service, generally on a monthly or annual per-user basis, and simply log in through a browser or a thin client application to start using the software immediately, with no local installation step at all.